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Post-Fault FOC Transformation Matrix for Unequal Amplitude Currents during Dual Open-Phase Fault

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Abstract—Many critical five-phase PMSM drive systems require operation after a dual open-phase fault has occurred. Previously documented techniques suggest that injecting unequal amplitude currents into the non-faulted phases can minimize the torque ripple generated by this type of fault. However, the only control method which can control these currents is a phase current hysteresis controller, which generates larger magnitude harmonic currents and provides less control of the transient response when compared to field-oriented control (FOC). Therefore, this paper proposes a novel method for developing a post-fault FOC transformation matrix which can produce constant transformed current references for a given set of three unequal amplitude phase currents. This transformation matrix is then implemented in an FOC architecture with two proportional-integral current controllers and a phase back-EMF feed-forward block. An example post-fault transformation is outlined in this paper for a set of unequal amplitude post-fault currents which are calculated using a previously documented post-fault optimization algorithm. This transformation is then implemented in an experimental setup to provide experimental validation that the proposed post-fault FOC technique can accurately control unequal amplitude phase currents with minimal distortion.

Keywords—five-phase, field-oriented control, transformation matrix

I. INTRODUCTION

Permanent-magnet synchronous machines (PMSMs) are being considered for many aerospace [1] and electric vehicle applications [2, 3]. These applications are safety-, mission- and cost- critical and hence do not tolerate system disruption. Therefore, it is beneficial that control solutions allow for safe machine operation, at a reduced output, after an electrical fault has occurred. There are many documented control techniques which have been developed for post-fault operation. Many of these techniques focus on five-phase systems, which provide more degrees of freedom in the post-fault optimization as compared to the three-phase system [4]. One important advantage of a five-phase system, is that it can continue to operate after a dual open-phase fault has occurred.

There are many documented post-fault control strategies for dual open-phase faults in a five-phase star-connected PMSM system. These methods determine the non-faulted phase currents which optimize the machine output torque. The optimization techniques presented in [5, 6] consider all the back-EMF and current harmonics in the algorithm but can generate high order current harmonic solutions, which present

practical control implementation issues. Hence, [4, 7, 8] only consider the first and third harmonic components of the back-EMFs and currents. The methods in [9, 10] generalize this approach for n -phases and for many types of stator winding configuration. Therefore, there are various documented techniques which optimize the machine output torque during a dual open-phase fault to a five-phase star-connected system.

The documented dual open-phase post-fault techniques can be grouped into three different approaches. The first approach only considers fundamental current solutions and constrains the three healthy phase currents to have equal magnitude. This maximizes the average post-fault torque per ampere. However, the neutral point forces the three post-fault current phasors to be 120° apart, which offers no degrees of freedom to minimize torque pulsation. Hence, this strategy is not beneficial as it generates significant torque ripple and outputs negative torque for part of the machine rotational cycle.

The second approach removes the equal current magnitude constraint to allow for degrees of freedom to minimize the torque ripple. This technique trades a reduction in the average torque per ampere for a considerable improvement in the torque ripple and the removal of the undesired negative torque periods. The third approach also includes unequal magnitude third harmonic phase currents into the optimization algorithm to obtain the smallest magnitude torque ripple. However, this current solution is more difficult to practically implement in a current control system than the previous two approaches.

All the post-fault control strategies, mentioned thus far, use phase current hysteresis controllers to control the desired post-fault currents. However, hysteresis controllers produce higher current harmonics, a wider frequency range of electromagnetic interference and provide less control of the transient response when compared to the field-oriented control (FOC) approach [11]. During normal steady state operation, FOC transforms the desired phase currents into constant current references, via a transformation matrix [12], and inputs these constant transformed currents into proportional-integral (PI) controllers. This means that the integral controller can reduce the steady state error to zero, which avoids obtaining tracking errors that occur when using phase current hysteresis controllers. Note that these tracking errors create undesired current harmonics and torque ripple and hence should be avoided. Therefore, a post-fault control strategy which uses FOC during a dual open-phase fault to a five-phase PMSM system is beneficial.

The standard FOC transformation matrix only produces constant steady state current references with the normal operating currents. Therefore, a new FOC transformation matrix is required for the dual open-phase fault and associated set of post-fault optimized phase currents. A new four-phase post-fault FOC transformation matrix is presented in [11] for a single open-phase fault to a five-phase system. However, there is currently no documented post-fault FOC transformation matrix for the dual open-phase fault. The standard three-phase FOC transformation can be used during this fault, but this means that the post-fault current phasors must have equal magnitude and must be 120° apart to avoid obtaining oscillating transformed reference currents which cause tracking errors. However, there is currently no reported post-fault FOC transformation matrix for generating constant current references with post-fault currents which have uneven amplitudes and different phase-shift angles during a dual open-phase fault to a five-phase system. This paper rectifies this by developing a generalized post-fault three-dimensional FOC transformation matrix which can be used to obtain constant reference currents, during steady state operation, with uneven amplitude phase current solutions. The new post-fault transformation matrix is then combined with PI current controllers to accurately control these post-fault currents with minimal distortion.

II. FOC TRANSFORMATION MATRIX THEORY

A. Standard Three-Phase FOC Transformation

This section of the paper outlines the standard three-phase FOC transformation, which is used in normal three-phase drive systems and outlines the associated transformed system equations. Note that the equations in this section are well-known and are not novel.

The aim of an FOC transformation matrix is to convert the reference phase currents to constant values, during steady state operation. To do this, the standard three-phase stationary transformation (1) maps the phase domain currents onto the α - β -0 axes. Under standard operation, these phase domain currents are balanced and hence yield zero zero-sequence current and a circular locus in the α - β plane. The circular rotational transformation (2) is then applied to these α - β transformed currents to obtain constant transformed reference currents in the d - q -0 axes. Therefore, the resultant standard FOC transformation (4) is calculated using (3).

The PI current controllers have d - q input current references and hence must compensate for the transformed voltage machine equation (5), where the transformed stator flux-linkage term λ_{trans} is (6). Note that (5) is the generic transformed voltage machine equation which holds true for any FOC transformation. The standard three-phase transformation matrix (4) contains the same phase-shift angles as the phase self- and mutual- inductance harmonics and the permanent magnet (PM) flux-linkage harmonics for a three-phase PMSM system. Therefore, the transformed inductance and flux-linkage components are (7-8) for a standard three-phase PMSM system and only possess constant values. The standard three-phase transformation matrix possesses the property outlined in (9). Therefore, (5) can be simplified to obtain (10). Note that the

last term in (10) cross-couples the transformed axes which can cause significant harmonics in the d - q current responses. However, under standard operation with a three-phase PMSM system, the λ_{trans} components are all constants, (7-8), which are independent of rotor position. Hence minimal cross-coupling occurs between the transformed axes.

The transformed machine terminal power (11) is used to calculate the transformed machine output torque. Also, the standard three-phase transformation possesses the property in (12). Therefore, (11-12) are used to obtain (13), assuming negligible reluctance torque.

B. Generic Post-Fault Three-Dimensional Transformation

This section of the paper explains which of system equations, outlined for the standard three-phase transformation, hold true for a generic three-dimensional post-fault transformation. A post-fault FOC transformation matrix must be designed to give constant transformed current references with the desired non-faulted phase currents. These non-faulted phase currents will typically not have the same phase-shift angles as the machine inductance and PM flux-linkage harmonics. Therefore, during post-fault operation, the transformed PM flux-linkage and inductance terms are (14-15). Note that these terms have rotor-position dependent components which cross-couple the transformed axes in (10). This cross-coupling will cause undesired current harmonic generation from the PI current controllers. To reduce the magnitude of this current harmonic generation, the phase back-EMFs can be fed-forward such that the PI current controllers only compensate for the resistive-inductive (RL) terms in $\lambda_{trans,f}$ (i.e. the PI controllers no longer compensate for $\lambda_{PM,f,trans}$). Note that the inductance harmonics in $L_{trans,f}$ cannot be easily fed-forward as obtaining accurate self- and mutual- inductance waveforms is difficult due to the effects of magnetic saturation. Therefore, the post-fault transformed machine equation will have some small magnitude cross-coupling terms due to the transformed inductance harmonics.

Note that the machine equation form in (10) minimizes cross-coupling between the axes. Therefore, a generic post-fault transformation must also comply with (9) to ensure that the transformed machine equation in (10) holds true. A post-fault transformation matrix which produces constant transformed values for uneven amplitude phase currents will not comply with (12). This is because the post-fault matrix will not be perfectly orthogonal due to the uneven phase current amplitudes. Therefore, (13) cannot be used with the new post-fault transformation to estimate the machine output torque.

$$T_{stat} = \frac{2}{3} \begin{bmatrix} 1 & -0.5 & -0.5 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad (1)$$

$$C_{rot} = \begin{bmatrix} \cos(\omega t) & \sin(\omega t) & 0 \\ -\sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

$$T(\theta) = C_{rot} T_{stat} \quad (3)$$

$$T(\theta) = \frac{2}{3} \begin{bmatrix} \cos(\omega t) & \cos\left(\omega t - \frac{2\pi}{3}\right) & \cos\left(\omega t + \frac{2\pi}{3}\right) \\ -\sin(\omega t) & -\sin\left(\omega t - \frac{2\pi}{3}\right) & -\sin\left(\omega t + \frac{2\pi}{3}\right) \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \quad (4)$$

$$v_{trans} = Ri_{trans} + \frac{d}{dt}(\lambda_{trans}) - \frac{d}{dt}(T(\theta))T(\theta)^{-1}(\lambda_{trans}) \quad (5)$$

$$\lambda_{trans} = L_{trans} i_{trans} + \lambda_{PM,trans} \quad (6)$$

$$L_{trans} = T(\theta)L_{abc}T(\theta)^{-1} = \begin{bmatrix} L_d & 0 & 0 \\ 0 & L_q & 0 \\ 0 & 0 & L_0 \end{bmatrix} \quad (7)$$

$$\lambda_{PM,trans} = T(\theta)\lambda_{PM,abc} = \begin{bmatrix} \lambda_{PM,d} \\ 0 \\ 0 \end{bmatrix} \quad (8)$$

$$\frac{d}{dt}(T(\theta))T(\theta)^{-1} = -\omega \times = \begin{bmatrix} 0 & -\omega & 0 \\ \omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (9)$$

$$v_{trans} = Ri_{trans} + \frac{d}{dt}(\lambda_{trans}) + \omega \times \lambda_{trans} \quad (10)$$

$$P = v_{trans}^T (T(\theta)^{-1})^T T(\theta)^{-1} i_{trans} \quad (11)$$

$$T(\theta)^{-1} = kT(\theta)^T \quad (12)$$

$$T_e(t) = \frac{P}{2} k \lambda_{PM,d} i_q \quad (13)$$

$$L_{trans,f} = T_f(\theta)L_{abc}T_f(\theta)^{-1} = \begin{bmatrix} L_{df1} & L_{df2} & L_{df3} \\ L_{qf1} & L_{qf2} & L_{qf3} \\ L_{0f1} & L_{0f2} & L_{0f3} \end{bmatrix} \quad (14)$$

$$\lambda_{PMf,trans} = T_f(\theta)\lambda_{PM,abc} = \begin{bmatrix} \lambda_{PM,df} \\ \lambda_{PM,qf} \\ \lambda_{PM,0f} \end{bmatrix} \quad (15)$$

III. PROPOSED FOC TRANSFORMATION MATRIX

This section details the proposed method for developing a three-dimensional post-fault FOC transformation for a given

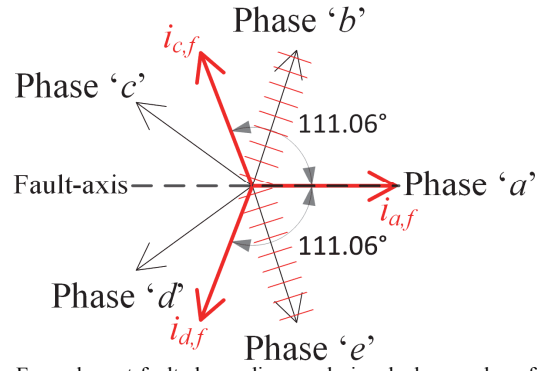


Fig. 1. Example post-fault phasor diagram during dual open-phase fault

set of three post-fault phase currents. The post-fault optimization strategy in [4] outlines that the post-fault currents during a dual-open phase fault should be (16-18). Note that currents i_2 and i_3 will possess phasors which are symmetrical about the faulted axis as shown in Fig. 1. Therefore, i_2 and i_3 have the same amplitude of $I_m \rho_1$ and same magnitude phase-shift angle γ_2 . The proposed post-fault FOC transformation matrix has been constructed for these currents in (16-18).

$$i_{1,f} = I_m \sin(\omega t + \gamma_1) \quad (16)$$

$$i_{2,f} = I_m \rho_1 \sin(\omega t + \gamma_2) \quad (17)$$

$$i_{3,f} = I_m \rho_1 \sin(\omega t - \gamma_2) \quad (18)$$

The post-fault stationary transformed currents must be (19) to ensure that the standard rotational matrix, (2), can be used to transform the stationary currents to constant reference current values. The proposed post-fault stationary transformation matrix is outlined in (20).

$$\begin{bmatrix} i_{\alpha,f} \\ i_{\beta,f} \\ i_{0,f} \end{bmatrix} = \begin{bmatrix} I_m \sin(\omega t) \\ -I_m \cos(\omega t) \\ 0 \end{bmatrix} \quad (19)$$

$$T_{stat} = \begin{bmatrix} u & v & w \\ x & y & z \\ 1 & 1 & 1 \end{bmatrix} \quad (20)$$

Therefore, two constraints (21-22) are established to ensure that the stationary currents in (19) are obtained from (20) with the reference phase currents (16-18). However, (21-22) contain rotor-position dependent terms which need to be removed.

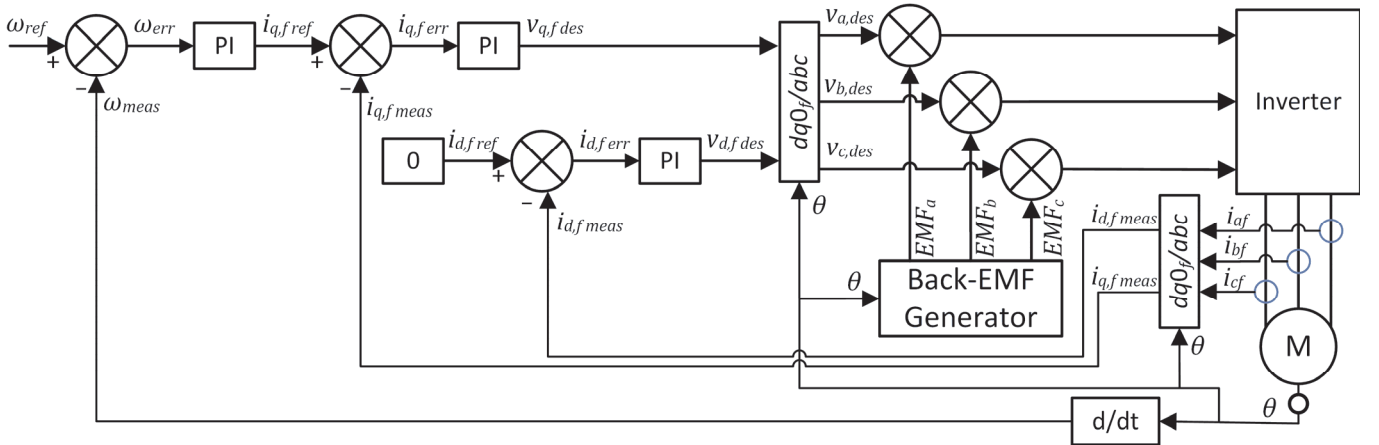


Fig. 2. Architecture of the post-fault FOC strategy with proposed transformation matrix during a dual open-phase fault to a five-phase PMSM system.

Hence, (21-22) are converted to exponential form and the imaginary and real components are grouped to obtain (23-26), which do not contain any rotor-position dependent terms.

$$u \sin(\omega t + \gamma_1) + v \rho_1 \sin(\omega t + \gamma_2) + w \rho_1 \sin(\omega t - \gamma_2) = \sin(\omega t) \quad (21)$$

$$x \sin(\omega t + \gamma_1) + y \rho_1 \sin(\omega t + \gamma_2) + z \rho_1 \sin(\omega t - \gamma_2) = -\cos(\omega t) \quad (22)$$

$$u \cos(\gamma_1) + \rho_1(v + w) \cos(\gamma_2) = 1 \quad (23)$$

$$u \sin(\gamma_1) + \rho_1(v - w) \sin(\gamma_2) = 0 \quad (24)$$

$$x \cos(\gamma_1) + \rho_1(y + z) \cos(\gamma_2) = 0 \quad (25)$$

$$x \sin(\gamma_1) + \rho_1(y - z) \sin(\gamma_2) = 1 \quad (26)$$

It is assumed that the post-fault rotor-position is aligned such that $\gamma_1 = 0$ to simplify (23-26). Also, the constraints (27-28) are added to ensure that the transformed α_f and β_f axes and the transformed β_f and 0_f axes are orthogonal. Note that there are not enough degrees of freedom to ensure that the transformed α_f and 0_f axes are orthogonal.

$$ux + vy + wz = 0 \quad (27)$$

$$x + y + z = 0 \quad (28)$$

The constraints (23-28) are solved to obtain (29-31), which completely define (20). Therefore, the proposed post-fault transformation matrix is determined using (20, 29-31) for a given set of post-fault currents (16-18). Note that these post-fault currents and ρ_1 and γ_2 can be established using the post-fault optimization method detailed in [4]. The proposed transformation matrix is then implemented in the FOC architecture shown in Fig. 2.

$$u = \frac{1}{(1 - \rho_1 \cos(\gamma_2))} \quad (29)$$

$$v = w = \frac{1}{2(\rho_1 \cos(\gamma_2) - 1)} \quad (30)$$

$$y = -z = \frac{1}{2\rho_1 \sin(\gamma_2)}, \quad x = 0 \quad (31)$$

IV. EXAMPLE POST-FAULT TRANSFORMATION

For the purposes of experimental verification, an example post-fault transformation is outlined in this section. This transformation has been determined for a non-adjacent dual open-phase fault to phases 'b' and 'e' of a five-phase system. The post-fault optimization algorithm in [4] has been used to calculate the optimal currents (32-34), which minimize machine torque pulsation for the PMSM specified in the Appendix. The proposed equations (20, 29-31) have been used to obtain the example post-fault stationary transformation (35). The example post-fault rotational transformation is then calculated online each algorithm cycle using (2-3, 35).

$$i_{a,f} = I_m \sin(\omega t) \quad (32)$$

$$i_{c,f} = 1.3912 I_m \sin(\omega t - 111.06^\circ) \quad (33)$$

$$i_{d,f} = 1.3912 I_m \sin(\omega t + 111.06^\circ) \quad (34)$$

$$T_{stat} = \begin{bmatrix} 0.6667 & 0.3333 & 0.3333 \\ 0 & 0.3851 & -0.3851 \\ 1 & 1 & 1 \end{bmatrix} \quad (35)$$

V. EXPERIMENTAL RESULTS

An experimental setup has been used to validate the example post-fault transformation detailed in section IV. This setup contains a dSPACE DS1006 processor and a DS5203 field-programmable gate array (FPGA) which implement the control algorithm and generate the IGBT gating signals respectively. The setup uses a custom-built five-phase two-level inverter which drives a prototype five-phase PMSM, with specifications outlined in the Appendix, and uses a 260 V dc bus. A dual open-phase fault to phases 'b' and 'e' fault has been simulated in this setup by removing these connections from the inverter terminals. The machine is then operated at a

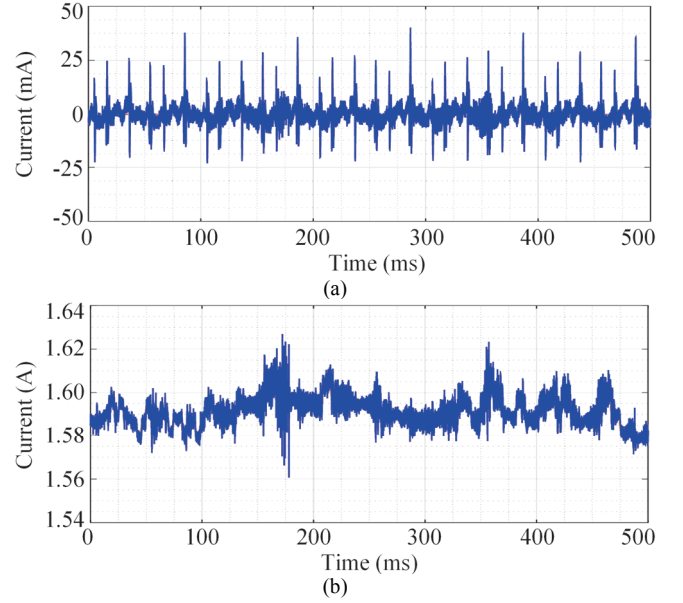


Fig. 3. Experimental transformed currents when operating with a rated current load torque at 300 rpm. Red line: Reference current, Blue line: Measured current. (a) $i_{d,f}$ current, (b) $i_{q,f}$ current.

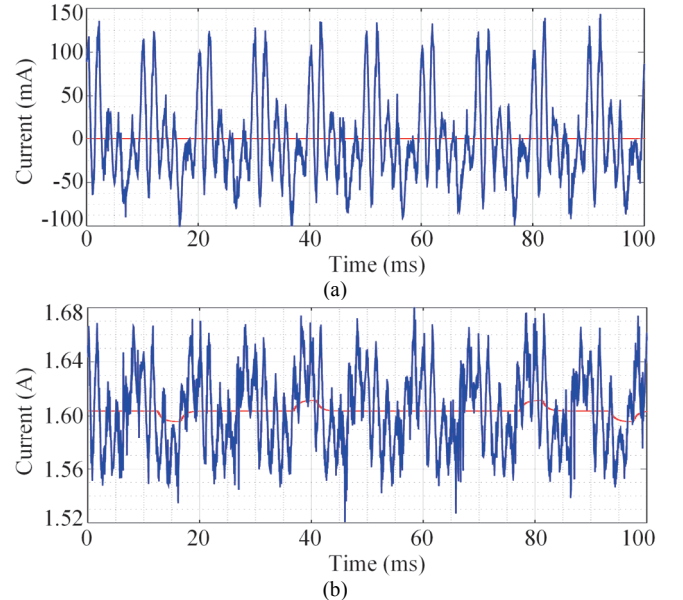


Fig. 4. Experimental transformed currents when operating with a rated current load torque at 1500 rpm. Red line: Reference current, Blue line: Measured current. (a) $i_{d,f}$ current, (b) $i_{q,f}$ current.

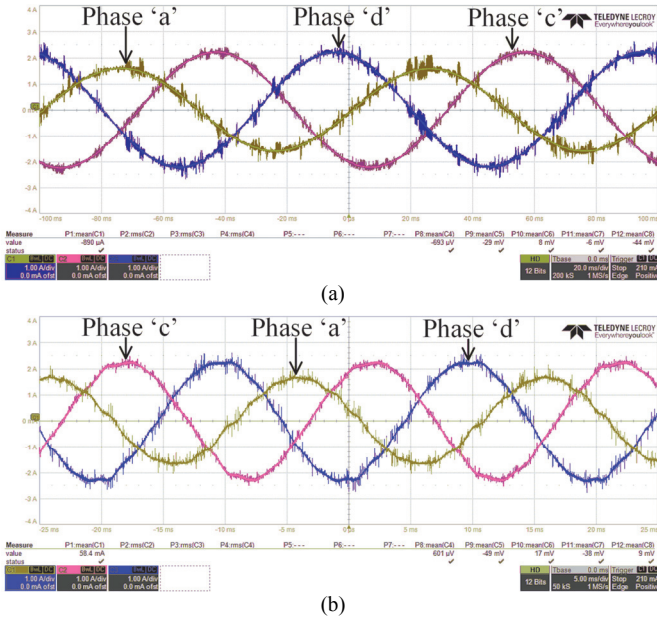


Fig. 5. Experimental measured phase currents. Current scale: 1 A/div. (a) Speed = 300 rpm (Time scale: 20 ms/div), (b) Speed = 1500rpm (Time scale: 5 ms/div).

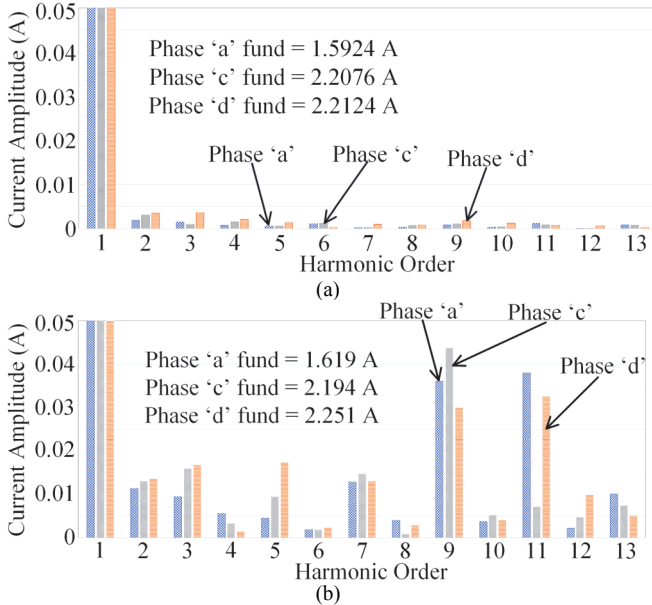


Fig. 6. Frequency spectra of measured phase currents. (a) Speed = 300 rpm, (b) Speed = 1500rpm.

TABLE I. TOTAL HARMONIC DISTORTION OF PHASE CURRENTS

Speed (rpm)	Total Harmonic Distortion (0 – 1000 Hz)		
	Phase 'a'	Phase 'c'	Phase 'd'
300	1.93%	1.89%	2.15%
1500	4.05%	2.67%	2.74%

low speed, 300 rpm, and a high speed, 1500 rpm, with the proposed example post-fault transformation and FOC architecture, Fig. 2, control technique. A load torque which requires the rated current from phase 'c' and 'd' has been applied to the machine for these tests. The results from these tests are shown in Figs. 3-6. This section will discuss these results.

In Fig. 3, the measured transformed i_d and i_q currents accurately track the reference currents at 300 rpm with approximately 65 mA and 50 mA of peak-to-peak current ripple respectively. At 1500 rpm, the measured transformed i_d and i_q currents still accurately track the reference currents but have an increased peak-to-peak ripple of approximately 250 mA and 160 mA respectively. Therefore, the magnitude of harmonic current generation increases as speed increases. This is because of the cross-coupling between the transformed axes in the machine equation caused by the last term in (10). This term possesses harmonics from all elements in L_{transf} (14), which cross-couple the axes. These components are multiplied by the machine rotational speed in (10), which is why increasing speed will cause increased cross-coupling between the axes.

The experimentally measured phase currents are sinusoidal in shape and possess negligible distortion at 300 rpm, Fig. 4. At 1500 rpm, these phase currents are still sinusoidal in shape but possess some small distortion due to cross-coupling. The frequency spectra of these phase currents, Fig. 5, shows that there are no significant undesired low-order harmonic components in the phase current response at 300 rpm. At 1500 rpm, Fig. 5 shows some smaller magnitude low-order harmonic components with amplitudes of less than 50 mA. The phase current total harmonic distortion (THD) values (for the 0 – 1000 Hz range) in Table I show that the maximum phase current THD is 2.15% at 300 rpm and 4.05% at 1500 rpm. These THD values are small, which means that the associated low-order harmonic currents will generate negligible torque ripple when compared to the developed average torque.

The phase-shift angles of the fundamental components in the frequency spectra, Table II, show that there is a maximum error of 0.4° from the desired angles at 300 rpm and a maximum error of 2° from the desired angles at 1500 rpm. The increase in phase-shift angle error at higher speeds is caused by the effects of cross-coupling.

The experimental torque waveforms in Fig. 6 show that there is approximately 0.24 Nm of peak-to-peak ripple with an average torque of 1.57 Nm at 300 rpm and 0.33 Nm of peak-to-peak ripple with an average torque of 1.58 Nm at 1500 rpm. Therefore, the peak-to-peak torque ripple increases from 15% of the average torque at 300 rpm to 21% of the average torque at 1500 rpm. This 7% increase in torque ripple at the higher speed is due to the small error increase in the fundamental current phase-shift angles and the small increase in phase current harmonic distortion, which are both due to cross-coupling. However, the previously documented phase current hysteresis current controller would have a significant tracking error at higher speeds, so would encounter a similar increase in torque ripple as speed increases. Therefore, this is not really considered a major limitation for this technique as all previous techniques face the same problem.

These experimental results validate the example post-fault transformation as they prove it can be used in an FOC architecture to accurately control three unequal amplitude phase currents with minimal distortion. This verifies the proposed method for developing a post-fault transformation for a given set of unequal amplitude currents during a dual open-

TABLE II. PHASE-SHIFT ANGLES OF FUNDAMENTAL CURRENTS

Speed (rpm)	Phase-Shift Angle (deg)		
	Phase 'a'	Phase 'c'	Phase 'd'
300	0°	-110.9°	111.3°
1500	0°	-109.0°	112.9°

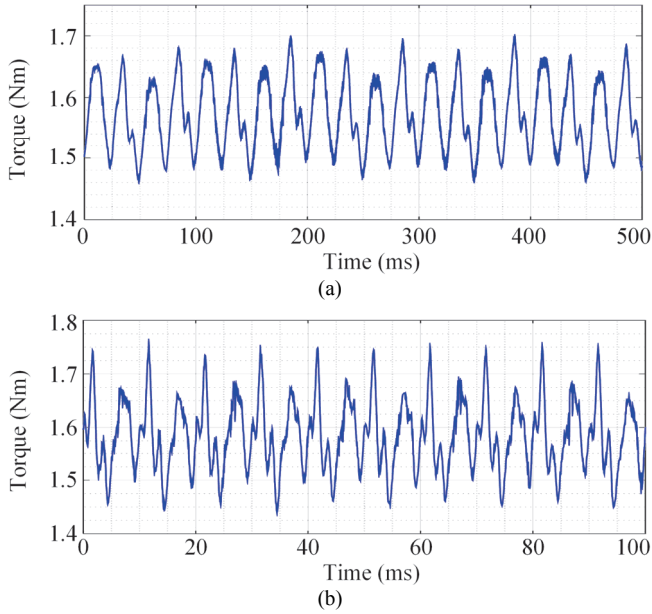


Fig. 7. Experimental torque waveforms when operated at rated current load. (a) 300 rpm, (b) 1500 rpm

phase fault to a five-phase star-connected system. The only limitation of the proposed technique is that slightly larger magnitude harmonic currents are generated at higher speeds. However, this is the only method which can accurately control three unequal amplitude phase currents in an FOC architecture.

VI. CONCLUSION

This paper proposes a novel method for developing a post-fault FOC transformation matrix (20, 29-31) which produces constant transformed reference currents with a given set of three unequal amplitude phase currents. The proposed transformation matrix is then used in an FOC architecture, in Fig. 2, to accurately control these unequal amplitude phase currents during a dual open-phase fault to a five-phase PMSM system. An example transformation matrix (35) is outlined based on three unequal phase currents (32-24), which have been calculated using the optimization algorithm presented in [4]. This example transformation is used to experimentally validate the proposed method. Experimental evidence confirms that the example post-fault technique can accurately control the phase currents with a maximum phase current THD of 2.15% at 300 rpm and 4.05% at 1500 rpm. These tests also show that the fundamental components of the measured phase current have a maximum error of 0.4° from the desired phase-shift angles at 300 rpm and a maximum error of 2° from the desired phase-shift angles at 1500 rpm. This is the only method which can accurately control three unequal amplitude phase currents in an FOC architecture.

APPENDIX

TABLE III. FIVE-PHASE PMSM SPECIFICATIONS

Freq. (Hz)	Back-EMF Voltage (V)	Self-Inductance Harmonics (mH)	Mutual Inductance Harmonics (mH)	
			Adjacent	Non-Adj.
dc	0	24.66	7.86	2.57
50	93.16	0	0	0
100	0	2.91	1.34	0.07
150	8.34	0	0	0
200	0	1.62	0.45	0.52
250	1.16	0	0	0
300	0	0.21	0.22	0.10
350	0.4	0	0	0

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